

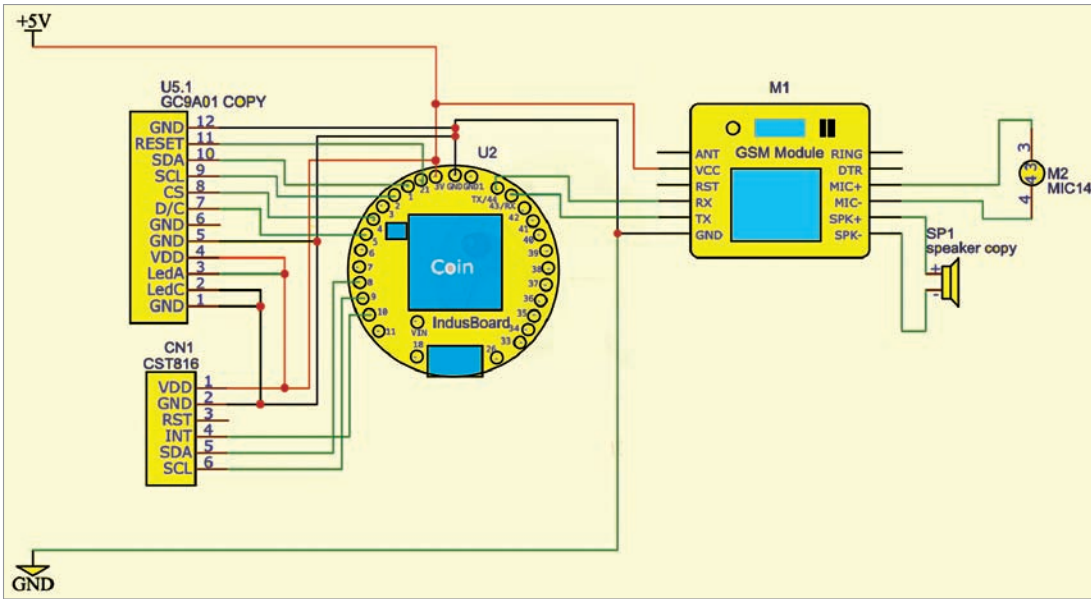


Fig. 9: IndusPhone circuit diagram

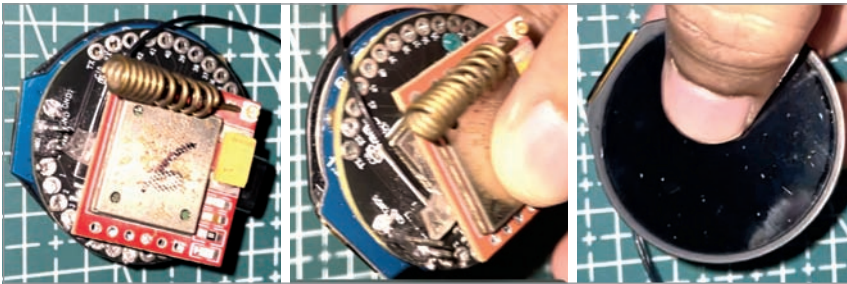


Fig. 10: IndusPhone components placement and soldering



Fig. 11: Typing message and dialling number

the display itself works on SPI, and the GSM SIM card module functions on serial. On the IndusBoard coin, any I/O pin can be designated for peripheral functions by defining the pin in the code. In this case, pins 8 and 9 are used for I2C (for the display), pins 1-7 for SPI, and pins 44 and 43 for

serial (for the GSM SIM module). After connecting and soldering all the components, fit them behind the IndusBoard coin, as shown in Fig. 10. Fig. 10 shows the IndusPhone components' placement and soldering.

Once the device is assembled, it can be powered and tested. Each function, including the GSM SIM and the UI for typing messages, dialling numbers, and attending calls, should be tested. Fig. 11 shows the process of typing a message and dialling a number.

EFY note. This is the first part of designing the world's smallest phone. Further parts will be published in a series, as the design is extensive and advanced. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/creating-smallest-wearable-phone-with-4g-volte-support-part-1>

Construction and testing

After gathering the components, solder them according to the circuit diagram. The display has a touch driver that operates on I2C, while

EFY Note
The source code
of this project
is available
for free download at
www.electronicsforu.com

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